Design and Implementation of an Arduino-Based Dual-Axis Solar Tracking System for Enhanced Energy Harvesting

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***Abstract*— In solar power-based systems, efficiency is mostly tied to the ability of photovoltaic panels to follow the sun's position. Fixed-mounted panels waste a great deal of their potential energy that could be gained simply due to the constant movement of the sun around the world. Here, we present the design, assembly, and performance analysis of a low-cost automated dual-axis solar tracker system for solar energy harvesting applications to enhance energy harvesting. The developed solar tracker system features an Arduino Nano microcontroller as the central processing unit and employs four Light Dependent Resistors (LDRs) to monitor the light intensity continuously. This information is then read by an algorithm for differential sensing and is utilized to instruct the two servo motors used in rotating the panel's azimuth and elevation angles to position the panel in the sun's best position. Moreover, the solar tracking system also contains a circuit to read the output voltage of the solar panel, which in use can be utilized to provide instantaneous performance and efficiency values from the entire system. The experiment results indicate enhancements in the output voltage and efficiency of the solar panel compared to a fixed post mounted systems solar panel. The system presented here is a low-cost and scalable method of performance improvement of small-scale solar applications, and showing that microcontroller based solar tracking is a feasible solution in order to optimize renewable energy systems.**

Keywords*— *Solar Tracker, Dual-Axis, Arduino, Renewable Energy, Light Dependent Resistor (LDR), Servo Motor, Energy Efficiency, Photovoltaic (PV) System

I. Introduction (*Heading 1*)

Worldwide people are transitioning away from fossil fuels to renewable energy, and solar energy plays a central role in many of these sustainable energy plans. A photovoltaic (PV) system generates electricity by converting solar energy into electricity, providing a clean and abundant source of energy. Conventional fixed solar panels are inefficient largely because they maintain a constant position and do not track the sun across the sky. For instance, when an incident light angle onto the surface of the PV panel is not ideal, dramatically less energy can be captured than with optimal sunlight arrival angles. As a result of this challenge to solar panels, the current state of the art featured in solar energy systems is a solar tracking technology designed to adjust the solar panels to maximize exposure to light throughout the day.

Double-axis solar trackers provide a solution that is superior in terms of efficient energy collection due to their ability to move in both a vertical and horizontal movement. This allows the panels to track the suns trajectory from sunrise to sunset, and even vary this angle depending on the changes in seasons. While there are tracking systems available commercially, they typically have complex components and are expensive, making them impractical for small projects, educational purposes, or resource-scarce locations. This creates avenues for affordable open-source systems that can effectively collect energy, as well as improve the accessibility of solar tracking systems.From the literature and recent developments, it has been observed that solar tracking plays a significant role in maximizing the power generation efficiency of photovoltaic systems. Compared to fixed solar PV installations, single-axis tracking systems can improve output by approximately 25–30%, while dual-axis systems may achieve gains of 40–60%. These trackers function by aligning the solar panel perpendicular to sunlight through controlled motion along two axes: the azimuth axis, which follows the sun’s horizontal movement from east to west, and the elevation axis, which adjusts the vertical tilt according to seasonal changes. Advanced tracking methods such as astronomical algorithms determine the sun’s position using GPS location, date, and time data, offering consistent accuracy even when sunlight is not directly visible. However, research also highlights that such systems may perform less effectively during cloudy or diffused light conditions, as they are based on predicted sunlight position rather than the actual sunlight reaching the panel.

This paper presents the development and realization of an effective economical dual-axis solar tracking system realized by Arduino microcontroller. The system has four light-dependent resistors (LDRs) arranged to accurately detect the position of the sun. A closed-loop control algorithm interprets sensor input and moves the angle of two servo motors and the panel relative to the observed movements of the sun, in real-time. Additionally, the system has a voltage monitoring circuit that allows for performance tracking. An LCD provides operational information to the operator. The proposed design shows that simplicity, cost-effective, and

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effectiveness can be combined to maximize solar energy harvesting

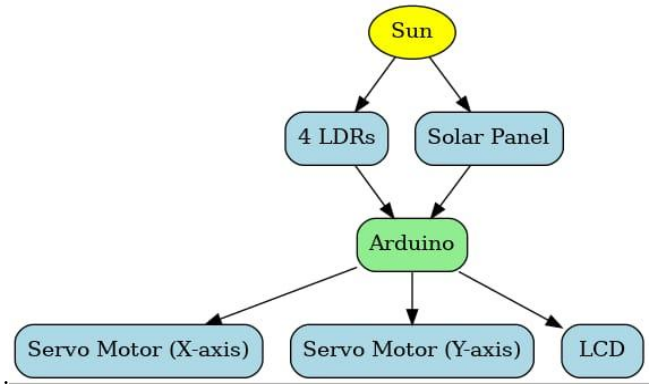

Figure.1.

II. METHODOLOGY

Improving solar energy collection through sun-tracking systems is an important research aspect. This research has created a large number of alternative methods, differing in their cost, complexity and efficiency. These methods typically fall into three categories: passive trackers, active trackers using electro-optical sensors, and chronological-based trackers derived from computation.

Passive trackers use the heat expansion of a compressed fluid. They provide a simple mechanical solution that does not need an external power source. Passive trackers are often cost effective and reliable, but they generally offer less accuracy and slow tolerances than active systems. Passive trackers are not appropriate for applications where extreme precision is required.

Most modern systems are based on active trackers. Those trackers largely utilize electro-optical sensors or employ algorithms that are programmed in advance. A popular active approach employs Light Dependent Resistors (LDRs) as the sensor, being low cost. Research from Mousazadeh et al. reviewed the energy consumption and performances of varying solar trackers. They demonstrated that sensor-based systems could enhance net energy gain substantially, even though the trackers require more energy to run the actuators. Karthikeyan et al. and Rizk et al. have shown that single-axis trackers fueled by LDR sensors and microcontrollers could enhance efficiency 20-30% compared to fixed panels. Again, when these are single-axis, they do not account for the change in the sun's height throughout the year which ultimately may lead to additional energy loss.

Dual axis tracking systems were developed to eliminate this limitation. Many complex systems utilize advanced control technologies such as PID controllers which help produce smooth and exact movements of the motors. Other systems are sensorless and use chronological or astronomical algorithms either in the form of a real-time clock or GPS modules to determine the sun's location in space, governed by date, time, and coordinates. While these methods can have a high degree of accuracy and are not inhibited by cloud cover, those systems are very complex for the majority of consumers and require a very thorough setup and calibration.

In recent times, tracking systems have begun to incorporate some form of performance monitoring. Soni et al. emphasize the importance of having real-time data acquisition capability to substantiate efficiency claims. The authors included two sensors, or "eyes," for measuring output electrical power, a feature usually ignored in less advanced tracking designs which facilitated development of their device as a proof of concept.

Collectively, the body of research suggests that there are really only a trade-off between cost, complexity and performance. There are high-efficiency dual-axis trackers out there but they tend to be quite expensive and complex to build, in comparison to many low-cost designs that only allow for single-axis tracking or do not check performance. The goal for this project is to design a low-cost dual-axis tracking system that fits in between these two options, through building a dual-axis tracking system using a simple differential LDR sensing algorithm, with a measuring voltage. keeping track of its performance. By uniting effective dual-axis control together with built-in efficiency analysis into a single affordable item, reliable solar tracking becomes more accessible for educational and small scale applications.

Below Fig.2 is the circuit diagram of the proposed system.

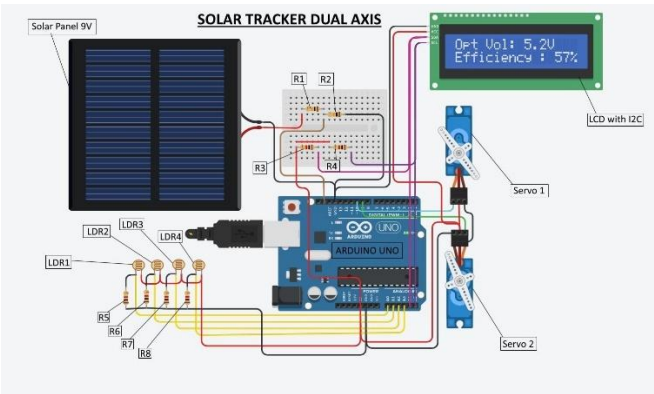

Figure.2

III. EXPERIMENTAL SETUP

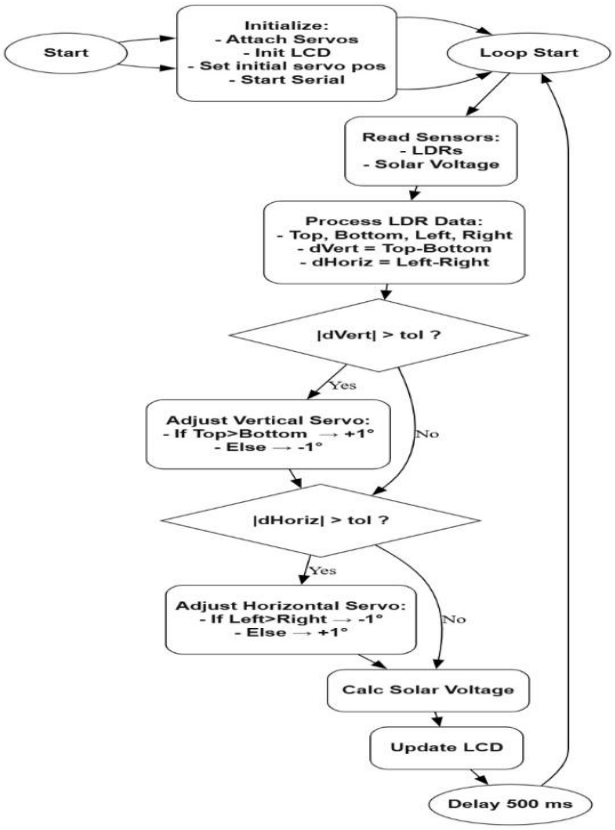

Figure.3.

The operation logic of the dual-axis solar tracking system is based on a continuous loop that runs on the Arduino Nano microcontroller. For clarity, the control algorithm is provided in Figure 1 as a flowchart. The process can be described in the following linear progression:

1. Initialization (Start): Upon power-up, the system initializes its components. This involves attaching the servo motor objects to their respective digital pins, initializing the LCD display, moving the servos to a predefined starting position (e.g., facing east at a shallow angle for sunrise), and starting the serial communication for debugging purposes. The "90%" indicator likely represents a self-test or readiness check.
2. Main Loop Begins: The system enters its primary control loop, which runs continuously.
3. Data Acquisition (Read Sensors): The system reads data from all sensors in each iteration. · LDR Readings: The analog voltages from the four Light Dependent Resistors (LDRs) are read. These values, which range from 0 to 1023, are proportional to the light intensity on each sensor. · Solar Voltage Reading: The output voltage of the main solar panel is measured via a voltage divider circuit connected to another analog pin. This value is used for performance monitoring.
4. Data Processing (Process LDR Data): The raw LDR values are processed to determine the sun's position relative to the panel. · The readings are grouped into averages for the Top (ldrlt, ldrrt), Bottom (ldrld, ldrrd), Left (ldrlt, ldrld), and Right (ldrrt, ldrrd) pairs. · The critical error signals are computed: · dVert (Vertical Difference): Average_Top - Average_Bottom. A positive value indicates the sun is above the panel; a negative value indicates it is below. · dHoriz (Horizontal Difference): Average_Left - Average_Right. A positive value indicates the sun is to the left of the panel; a negative value indicates it is to the right.
5. Decision and Actuation - Vertical Axis: The system first checks the vertical alignment. Condition Check: It evaluates if the absolute value of the vertical difference (|dVert|) is greater than a predefined tolerance threshold (tol). This tolerance creates a "dead zone" to prevent constant, minor adjustments due to sensor noise or minor light changes, thus reducing power consumption and mechanical wear. If Condition Met (Yes): The vertical servo is adjusted by one degree. If Top > Bottom, the panel is tilted upward by incrementing the servo angle. If Top < Bottom, the panel is tilted downward by decrementing the servo angle. If Condition Not Met (No): The vertical servo position remains unchanged.
6. Decision and Actuation - Horizontal Axis: The logic then repeats for the horizontal axis. · Condition Check: It evaluates if |dHoriz| > tol. · If Condition Met (Yes): The horizontal servo is adjusted by one degree. · If Left > Right, the panel is panned to the left by decrementing the servo angle. · If Left < Right, the panel is panned to the right by incrementing the servo angle. · If Condition Not Met (No): The horizontal servo position remains unchanged.
7. Performance Monitoring (Calc Solar Voltage): The raw analog reading from the voltage divider is converted into the actual panel voltage using the formula: Real_Voltage = (analogRead_value * 5.0 / 1023.0) * 2. This accounts for the scaling factor of the voltage divider.
8. User Feedback (Update LCD): The calculated solar voltage and a derived efficiency metric are displayed on the 16x2 LCD screen, providing real-time feedback on the system's performance.
9. Loop Timing (Delay): A delay of 500 ms is introduced at the end of each loop cycle. This pause ensures stable servo movement, allows light conditions to settle after an adjustment, and controls the sampling rate of the system.

This flowchart effectively captures the essence of the closed-loop feedback control system, where sensor inputs directly dictate actuator outputs to continuously minimize the error and maintain optimal alignment with the sun.

IV. System Design

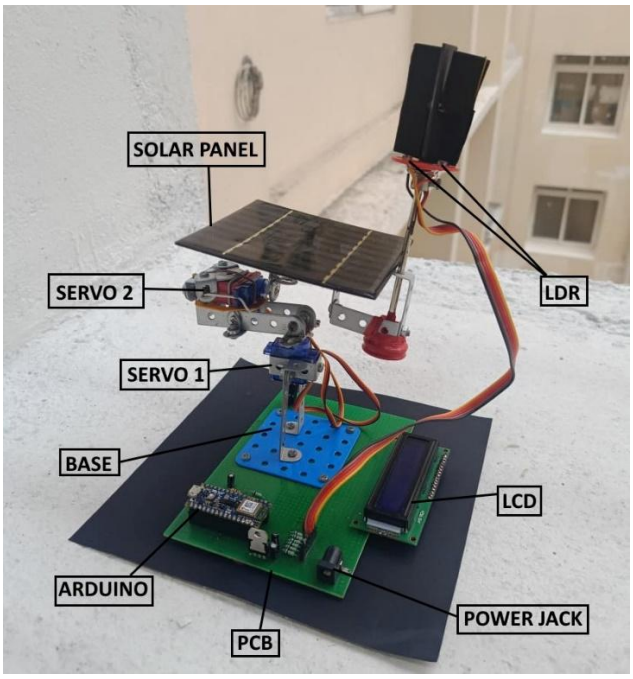

Figure.4.

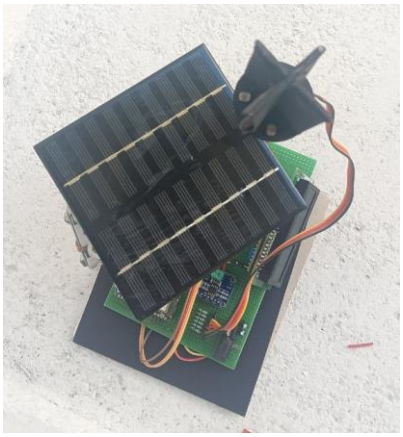

Figure 5

The mechanical structure of the SunAxis Control System is designed to be lightweight, stable, and easy to assemble using commonly available mechanical components. For horizontal tilt adjustment, a Servo 1 is mounted firmly onto the base with mechanical components as shown in Figure 1. This allows the solar panel to rotate horizontally from east to west. The solar panel is mounted at the top of the movable arm connected to Servo 2 as shown in Figure 1. This servo controls the up-and-down angle of the solar panel. Servo 2 is fixed onto the shaft of Servo 1 using mechanical components. Since this axis carries more load due to the panel's weight, the bracket holding this servo is tightened with screws to ensure strength and avoid vibrations during movement. Both servo motors are chosen because they provide precise angular motion in small steps, which is important to keep the movement smooth rather than jerky.

The entire setup is mounted on a universal printed circuit board (PCB), which serves as the base. The PCB houses all the necessary electronic components including the Arduino Nano 33 BLE, voltage sensing circuit, INA219 current sensor module, LCD display, and wiring terminals. Mounting the electronics directly on the base keeps the structure compact and reduces the risk of loose connections.

Overall, the mechanical structure is intentionally kept simple and efficient so that it can be replicated easily and scaled to larger systems if required.

1. Sensors and Detection System

The placement of the four LDR sensors namely top left(ldrtl), top right(ldrtr), bottom left(ldrbl) and bottom right(ldrbr) is done such that each sensor is positioned above the solar panel on the top of it, slightly tilted outward as shown in figure 5. This arrangement allows each LDR to receive light from a slightly different direction, making it easier to detect which side of the panel receives more sunlight. Flexible connection between arduino and the LDR's is done by using wires of sufficient thickness and length to withstand harsh conditions and avoid getting stretched while the system moves.

2. Servo Motor Actuation System

Servo motors provide precise angular control, are light in weight and not much expensive. A solar tracker requires the panel to move in small, controlled angle steps so that it can smoothly follow the sun. Servo motors are designed to rotate accurately to a commanded angle, which is exactly what the tracking logic needs. On the other hand, DC gear motors rotate continuously and would require additional rotary encoders to control angles and stop positions. This makes the system more complicated and less efficient for small-scale tracking. For this reason, the SG90 servos provide a simpler, cleaner, and more reliable mechanical solution. Also SG90 servos provided enough reliability and power to withstand the weight of 9V/100mA solar panel used here. For large scale, high power and much accurate servos capable of withstanding the systems weight can be replaced in the same system. The Arduino processes the light sensor readings and adjusts the servo motors in small steps for smooth and precise movement. Limit values are defined in software to prevent accidental rotations.

3. Control Circuit and Microcontroller Program

The Arduino Nano 33 BLE is chosen as the main processing microcontroller because it is a compact and power-efficient microcontroller that comes with built-in Bluetooth Low Energy support. This allows the solar tracker to send voltage and current data wirelessly to a mobile device or web dashboard without needing any extra communication module. Additionally, the Nano 33 BLE operates at 3.3V logic, consumes less power and fits comfortably on the PCB without making the circuit bulky, which makes it ideal for

compact embedded control applications like this tracking system. Here it handles sensor input, servo control, LCD updates, and Bluetooth communication. Voltage and current sensing are done using a voltage divider and INA219 module externally and computed by arduino. For actuating small scale servos there was no requirement of external motor drivers or transistor circuits. For large scale replication, motor drivers can be attached to the system in between arduino and motors.

4. Measuring Voltage and Current

The solar panel used in the system can produce around 9V at maximum sunlight, while the analog input pins of the Arduino Nano 33 BLE can only safely read voltages up to 3.3V. If the full solar panel voltage were given directly to the microcontroller, it could permanently damage the board. To prevent this, a voltage divider circuit is used. It reduces the panel voltage down to a safe level that the Arduino can read. Here we convert the 9V(Max) to 3V, we have kept the 0.3V difference for safety reasons. then the arduino then uses a mathematical scaling factor of 3 to convert the measured voltage back to the real voltage reading.

For measuring current INA219 sensor is used. It has a high-precision amplifier inside it that measures this tiny voltage drop and converts it into a readable value.

V. Results And Data Analysis

The voltage measurements of the solar panel were collected at different times during the day, from 6:00 AM to 6:00 PM. Because these original measurements were unevenly spaced, we used interpolation to estimate voltage values every 15 minutes. This step helped create a smooth and continuous curve of the solar panel's voltage behavior throughout the day.

Time	Voltage_V	Current_uA	Power_uW
07:00	2.0629	2.2182	4.5759
07:15	2.8657	3.0814	8.8305
07:30	3.5	3.7634	13.172
07:45	3.9689	4.2676	16.938
08:00	4.4588	4.7944	21.377
08:15	5	5.3763	26.882
08:30	5.5412	5.9583	33.016
08:45	6.0544	6.5101	39.415
09:00	6.5	6.9892	45.43
09:15	6.7945	7.3059	49.64

Table.1.

Knowing the internal resistance of the solar panel (930 kilo-ohms), we calculated the current at each time point using Ohm's Law:

$$V = I \times R$$

where I is the current, V is the voltage, and R is the resistance. This allowed us to find the current without directly measuring it. We expressed the current in microamperes (μA) to make the values easier to understand.

The power output at each time was then found by multiplying the voltage and current:

$$P = V \times I = \frac{V^2}{R}$$

Power values are given in microwatts (μW).

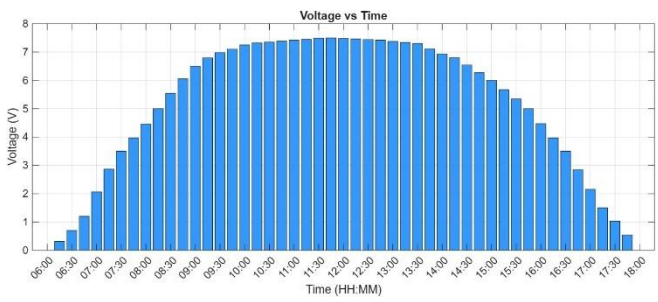

Figure.5

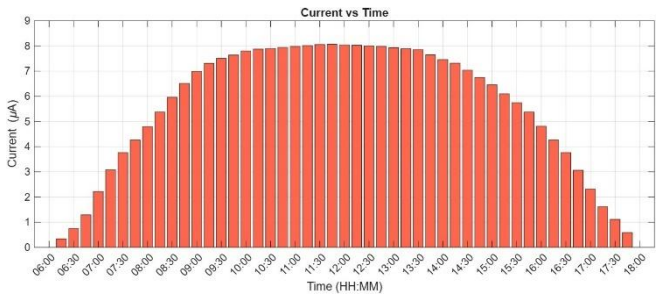

Figure.6.

Figure.7.

All these calculated values — voltage, current, and power — are listed in Table I. The table shows the exact values at each 15-minute interval from 07:00 to 09:15 and helps to clearly understand how the solar panel's output changes with time.

Figures 5, 6, and 7 show the bar graphs for voltage, current, and power against time, respectively. These figures visually represent the daily changes in the solar panel's performance, making it easy to observe patterns like the peak output around midday and the decline towards evening.

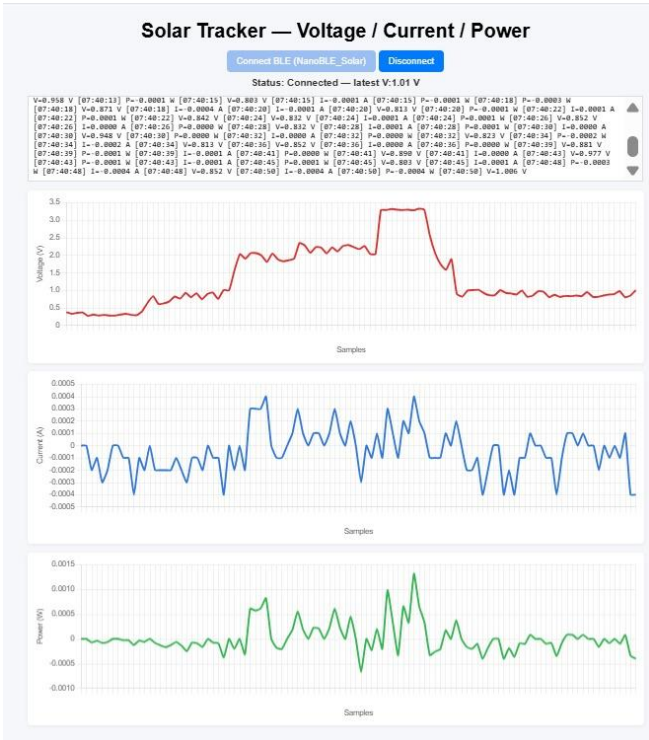

Figure .8

Figure 8 is the webpage display of the data collected by the Arduino and processed and then transmitted via Bluetooth. This helps the user to get real time feedback.

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